

Characteristics of a Semiconductor Laser Coupled With a Fiber Bragg Grating With Arbitrary Amount of Feedback

Alexander Naumenko, Pascal Besnard, Natalia Loiko, G. Ughetto, and J. C. Bertreux

Abstract—Stationary characteristics (such as laser frequency at threshold, light-intensity curve, etc.) and relative intensity noise of laser diodes coupled with an external Bragg reflector are studied for arbitrary amount of optical feedback. We put the stress on the interplay between external and internal cavities when the feedback strength is increased. For strong feedback, new stationary solutions that do not exist in the case of conventional feedback appear, while the ellipse, which usually contains the modes and so-called anti-modes, collapse in the plane frequency-gain. Finally, analytical expressions are compared with numerical results obtained through a new approach that uses Green functions.

Index Terms—Fiber Bragg grating (FBG), relative intensity noise (RIN), semiconductor laser.

I. INTRODUCTION

SEMICONDUCTOR lasers are extremely sensitive to optical feedback [1]–[5]. This external perturbation may lead to modifications of light-intensity curves and to a variety of dynamical and spectral behaviors such as mode hopping or the so-called “coherence collapse.” This last phenomenon is characterized by a catastrophic enhancement of the laser linewidth. These different regimes depend on two main parameters: the feedback strength and the time delay. A classification was introduced by Tkach [2] and five regimes were identified. Among them, in the third one (III), the coupled laser is operating stably on the external cavity mode which has the narrowest linewidth.

A more stringent technique to maintain single-mode operation consists in introducing a frequency-selective feedback. Several filtering schemes were realized by using a narrow-band resonator [6]–[8], a grating [9], [10], or a fiber Bragg grating (FBG) reflector [11]–[13]. One purpose is to obtain highly coherent sources for metrology [14] or sources for optical telecommunication [15]. The idea is to use frequency-selective elements with generally strong optical feedback obtained by an antireflection coating of the coupling laser face. It makes it possible to gain in coherence or to fix a specific wavelength. These types of cavities can be used to generate mode-locked lasers by gain commutation [16].

Theoretical investigations of such systems, in which a FBG was considered as a frequency-selective element, were undertaken in [17]. For high feedback levels, multiple reflections in the external cavity (EC) were taken into account. This inclusion is necessary in the case of strong feedback as it was emphasized earlier [1], [18]–[20] for frequency-independent external reflector. Spectral characteristics and relative intensity noise (RIN) as well as bistability and subharmonic generation have been studied with such terms [21]–[23]. Thus, for a very high quality factor, it is important to include a multiple round-trip approach for an accurate determination of steady states and bifurcation points as well as for numerical calculations of dynamics. Note that analytical investigations for an FBG-EC laser under strong feedback conditions were performed in the small-signal modulation regime using a Volterra functional series expansion method [17]. In the present study, we decided to adopt a multiple round-trip approach, whatever the feedback strength, in order to have general expressions (especially analytical ones). In the numerical simulations, we check that a single round trip can be generally considered for a bad quality factor.

In this paper, we expand our preliminary investigations [24] and present detailed studies of steady states and numerical simulations of such coupled systems. We use a model similar to the one presented in [17]. But in comparison with [17], the present study is not only analytical. It considers different types of FBG: uniform FBGs, FBGs characterized by a cosine envelope of the refraction-index profile (that is called cosine apodization [25]), etc.. In fact, any kind of filter may be incorporated.

Our model is detailed in Section II. To analyze changes in laser characteristics from weak to very strong feedback strength, by a unified way, we choose to take into account the EC through a transfer function which is written in the frequency domain and which includes multiple round trips. For numerical simulations, we introduce a new method which uses Green functions. Finally, we consider the multivalued character of the function describing the photon decay rate in the compound cavity. This allows us to study different internal cavity resonances while the laser is considered as a single-longitudinal mode device. As a matter of fact, we demonstrate that this approach which considers a monochromatic spectrum for the internal cavity (IC) can give results similar to those obtained through the use of the full set of Maxwell equations for the laser field [26].

Section III is devoted to the study of steady states and to their analysis. It is shown that under strong feedback conditions new branches of solutions appear in comparison to conventional

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feedback. They consist of EC modes and antinodes connected with other IC resonances. Moreover, we show that narrow filters lead to the excitation of additional solutions corresponding to side maxima in the FBG frequency response. However, they can be suppressed by using an apodized index grating.

In Section IV, we detail the dependence of the steady-state frequencies as a function of the solitary-laser frequency. The path among the steady states under slow pump current modulation are obtained and corresponding light–intensity (L – I) curves are described. The dependence of these characteristics on the amount of feedback is analyzed. Three regions of feedback strength are distinguished, showing a qualitatively different interplay between EC modes and IC ones. When the feedback strength is increased, the set of steady states is modified. This modification illustrates the way, EC modes are transformed into longitudinal modes of the long cavity composed of the laser and of the external mirror. In an intermediate region, jumps between solutions belonging to different IC resonances are observed, though the laser operation is kept at a frequency corresponding to that of the free laser. The possibility of IC multimode regime in this domain gives the limits of our model. In this case, an approach with several IC modes has to be introduced. It can be achieved by decomposition of the laser field into longitudinal modes [27], [28] or by using a spatio-temporal approach [26], [29].

In Section V, the dependence of RIN on different parameters is given and discussed for a typical diode coupled with FBG, used as a pump in EDFA. The results are closely linked to the existence of the three regions of feedback strength.

Finally, conclusions are drawn in Section VI.

II. EQUATIONS FOR AN ARBITRARY AMOUNT OF FILTERED FEEDBACK

A. Extended Cavity Approach

In our simulations, we consider the FBG laser system from the point of view of an extended cavity approach. The corresponding scheme is presented in Fig. 1. In this approach, the active medium is placed between two mirrors (which form the IC) for which the amplitude reflection coefficients are r_1 and r_2 . We assume that the solitary laser is a single-frequency device.

The EC is formed by the coupling mirror (reflectivity r_2) and an external Bragg mirror. Its reflectivity $r_B(\omega)$ is dependent on the laser frequency ω . As it was mentioned above for the case of strong optical feedback, it is necessary to take into account multiple round-trips of the optical field inside the EC. These assumptions enable us to write the following equations:

$$\begin{aligned} \dot{E} = & \left[\frac{1}{2} \Gamma G_N \{ i\alpha (N - N_{th}) + g(N, I) \} \right. \\ & \left. - \frac{1}{2\tau_{ph}} + \frac{1}{\tau_{in}} \ln \left(\frac{F}{E} \right) + i\Delta\omega_0 \right] E \\ \dot{N} = & \frac{J}{eV} - \frac{N}{\tau_N(N)} - G_N g(N, I) I. \end{aligned} \quad (1)$$

Here, $E(t)$ is the slowly varying complex amplitude of the laser field at time t . This amplitude can be written $E(t) = \sqrt{I(t)} \exp(i\varphi(t))$, where $I(t)$ is the laser field intensity and $\varphi(t)$ is the phase. The optical field is

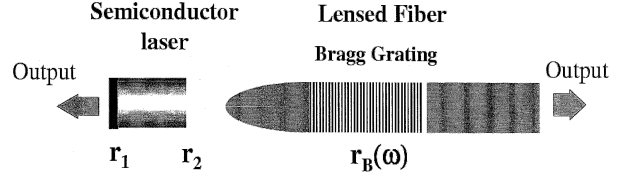


Fig. 1. Schematic view of a diode laser coupled with a FBG.

TABLE I
TABLE OF PARAMETERS

parameter	description
$V = 1.1 \times 10^{-16} \text{ m}^3$	volume of solitary laser
$G_N = 3 \times 10^{-20} \text{ m}^2$	differential gain
$N_0 = 1.5 \times 10^{24} \text{ m}^{-3}$	transparency carrier density
$\alpha_{in} = 800 \text{ m}^{-1}$	scattering losses
$\Gamma = 0.08$	field confinement factor
$\tau_e = 0.8 \text{ ns}$	spontaneous carrier lifetime
$J_0 = 0.0288 \text{ A}$	shift of the injection current
$R_1 = 0.9$	reflectivity of left facet
$\tau_{in} = 24 \text{ ps}$	cavity life time
$L_{ext} = 1 \text{ m}$	external cavity length
$n_{ext} = 1.5$	effective index of the fiber
$\alpha = 4$	linewidth enhancement factor
$\epsilon = 1.6 \cdot 10^{-19} \text{ sm}^3$	gain compression factor

$E_{opt}(t) = \text{Re}(E(t) \exp(i\omega_0 t))$, where ω_0 is the optical frequency of the solitary laser. The nonlinear gain is expressed as $g(N, I) = (N - N_0) / (1 + \epsilon I)$, where ϵ is the gain compression factor, $N(t)$ is the carrier density, with N_0 its value at transparency and N_{th} , at threshold $N_{th} = N_0 + 1/(\Gamma G_N \tau_{ph})$. G_N is the differential gain and Γ is the field confinement factor. The photon lifetime τ_{ph} is determined by the full losses of the solitary laser: $1/\tau_{ph} = \Gamma_{ph} = v_g \alpha_{in} - (1/\tau_{in}) \ln(R_1 R_2)$, where the parameter α_{in} determines the scattering losses in the active volume, $v_g = c/n_g$ is the group velocity, c is the speed of light in vacuum, n_g is the group index of the active medium. Parameters R_1 and R_2 are the power reflectivity of left and right facets: $R_{1,2} = r_{1,2}^2$. The cavity lifetime is $\tau_{in} = 2L_D/v_g$ and L_D is the laser diode length, α is the linewidth enhancement factor, J is the injection current, and e is the electron charge. The spontaneous emission rate $N/\tau_N(N) = AN + BN^2 + CN^3$ with A , B , and C the mono-molecular, bimolecular, and Auger recombination coefficients. Linearization of the spontaneous emission rate near N_{th} gives $N/\tau_N(N) = N/\tau_e + J_0/(eV)$, where τ_e is the carrier lifetime, V is the volume of the active zone, and J_0 is a shift of the injection current. $\Delta\omega_0$ is a frequency chirp due to thermal effects when the injected current is changed. The empirical dependence for this chirp is linear: $\Delta\omega_0 = k(J - J_{th})$. The influence of this term will keep our attention in a next paragraph. $\tau = 2L_{ext}n_{ext}/c$ is the EC round-trip time, where L_{ext} is the EC length and n_{ext} is the refraction index. The global parameters are given in Table I.

B. Effective External Reflectivity

The feedback terms corresponding to the influence of the external Bragg reflector in the multipass approach can be written as

$$F(t) = E(t) + \frac{(-1)(1-R_2)}{2\pi R_2} \times \sum_{n=1}^{\infty} \exp(-i\omega_0 n\tau) \int_{-\infty}^{+\infty} d\omega (-r_2 r_B(\omega))^n \times \exp(i\omega(t-n\tau)) \int_{-\infty}^{+\infty} dt' E(t') \exp(-i\omega t') \quad (2)$$

where n indicates the contribution of the n th EC round trip. The quantity $r_2 F(t)/E(t)$ can be interpreted as the temporal effective reflectivity of the EC and can be expressed in the frequency domain as $r_2 f(\omega)$, with

$$f(\omega) = 1 - \frac{1-R_2}{R_2} \sum_{n=1}^{\infty} \exp(-i(\omega_0 + \omega)n\tau) (-r_2 r_B(\omega))^n = 1 + \frac{1-R_2}{r_2} \frac{r_B(\omega) \exp(-i(\omega + \omega_0)\tau)}{1 + r_2 r_B(\omega) \exp(-i(\omega + \omega_0)\tau)} \quad (3)$$

where $f(\omega)$ and $F(t)$ are linked through the Fourier transform

$$F(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} d\omega f(\omega) \exp(i\omega t) \times \int_{-\infty}^{+\infty} dt' E(t') \exp(-i\omega t') \quad (4)$$

In numerical calculations, we will use the expression (2) for the feedback term $F(t)$ with separate contributions from EC round trips, as will be explained below.

C. Bragg Reflectivity

The complex reflectivity $r_B(\omega)$ of a uniform FBG is simply written with the help of the coupled-mode theory [25]

$$r_B(\omega) = \frac{-\varrho \sinh\left(L_B \sqrt{\varrho^2 - \delta^2}\right)}{\left(\sqrt{\varrho^2 - \delta^2} \cosh\left(L_B \sqrt{\varrho^2 - \delta^2}\right) + i\delta \sinh\left(L_B \sqrt{\varrho^2 - \delta^2}\right)\right)} \quad (5)$$

where ϱ is the strength of the FBG, L_B is the FBG length, and 2δ is the difference between the wave vectors of the incident and of the reflected fields. If n_B is the effective index of the FBG, Λ_B is the FBG period and λ_0 is the wavelength associated to the optical frequency $\omega_0 + \omega$, then $\delta = 2n_B\pi/\lambda_0 - \pi/\Lambda_B = (n_B/c)(\omega - \omega_B) = (n_B/c)\Delta\omega$. $\Delta\omega$ is the detuning from the Bragg frequency $\omega_0 + \omega_B$ associated to the Bragg wavelength $\lambda_B = 2n_B\Lambda_B$. Note that in our numerical simulations, the frequency is referred to ω_0 (only deviations from this frequency are considered). The reflectivity at the Bragg wavelength (null detuning) takes the usual form: $r_B(\omega_B) = \tanh(\varrho L_B) (\simeq \varrho L_B$ for small reflectivity). In further calculations we will use the following definitions for the coupling constant κ and the bandwidth Δ of the Bragg reflector $\kappa = \varrho L_B$, $\Delta = c/(L_B n_B) \sqrt{\pi^2 - \kappa^2}$, ($\simeq \pi c/(n_B L_B)$).

As mentioned above, we will not only consider a uniform FBG but also an index grating with some profile. This apodization is assumed to have zero “dc” index change, because in that case, the reflectivity curve does not show strong blue-side oscillations and is symmetric with respect to the central frequency. On the basis of the coupled-mode theory, the ratio $\rho(z)$ between the amplitudes of forward and backward traveling waves is described by a Ricatti differential equation which can be numerically integrated [30], [31]

$$\frac{d\rho}{dz} = i [2\delta\rho + \varrho\eta(z)(1 + \rho^2)] \quad (6)$$

Here, $\eta(z)$ is an apodization function: for “raised-cosine” apodization profile $\eta(z) = (1/2)[1 + \cos(2\pi z/L_B)]$ [25]. For a Gaussian profile truncated at three full-width-at-half-maximum $\eta(z) = \exp[-(az/L_B)^2]$, with $a = 6\sqrt{\log 2}$. Integration of (6) has to be performed from $-L_B/2$ to $L_B/2$ with the boundary condition $\rho(L_B/2) = 0$. Then the reflectivity of the FBG is defined as $r = (1/i)\rho(-L_B/2)$ and is maximum at zero detuning δ : $r_{\max} = \tanh(\varrho L_B \bar{\eta})$. The weight $\bar{\eta}$ is an averaged value of $\eta(z)$ and in the case of “raised-cosine” apodization profile, the weight is $\bar{\eta} \equiv 1/L_B \int_{-L_B/2}^{L_B/2} \eta(z) dz = 1/2$.

For low grating reflectivity level, (6) can be linearized [30] and the FBG reflectivity is given by the integral $r = -\varrho \exp(-i\delta L_B) \int_{-L_B/2}^{L_B/2} \eta(z) \exp(-i2\delta z) dz$. Explicit forms can be obtained for raised-cosine apodization

$$r = \frac{-\varrho L_B \exp(-i\delta L_B) \left(\frac{\sin(\delta L_B)}{(2\delta L_B)}\right) \pi^2}{\left(\pi^2 - (\delta L_B)^2\right)}$$

and for uniform grating

$$r = -\varrho L_B \exp(-i\delta L_B) \sin\left(\frac{\delta L_B}{\delta L_B}\right).$$

D. Green Functions for the Delayed Terms

The term $1/\tau_{\text{in}} \ln(F/E)$ in the field (1) is determined by the losses and phase conditions brought by the external mirror in the feedback loop. Usually this term is expanded in series and only the first term is taken into account [1]. This approximation is sufficient for a weak feedback, but as argued in the Introduction, we have to keep the logarithm as well as multi-pass approximation for F in case of strong feedback strength. As we will show below, this enables us to take into consideration different IC resonances along with EC resonances. However, no beating between these IC modes separated by $2\pi/\tau_{\text{in}}$ can be accepted in our system.

Moreover, note that the dependence of FBG reflectivity is in the frequency domain, while differential equations (1) for the laser field and the population inversion are given in the mixed frequency-time domain. For the sake of simplicity, we do not take into account frequency-dependent gain in the present investigation. This last assumption offers the possibility to do some simplification in order to calculate the feedback term $F(t)$ and to decrease the time of calculation.

Let us introduce the following notations:

$$G_n(x) = -\frac{1-R_2}{R_2} \exp(-i\omega_0 n\tau) \frac{1}{2\pi} \int_{-\infty}^{+\infty} d\omega (-r_2 r_B(\omega))^n \exp(i\omega x). \quad (7)$$

Then the contribution from the n th round trip in the sum (2) is given by the convolution product

$$\begin{aligned} F_n(t) &= G_n(t) \otimes E(t - n\tau) \\ &= \int_0^{+\infty} dx E(t - n\tau - x) G_n(x) \end{aligned} \quad (8)$$

and the whole term describing the FBG feedback is

$$F(t) = E(t) + \sum_{n=1}^{\infty} F_n(t). \quad (9)$$

Here, $G_n(x)$ represents the n th round-trip time response of the EC to a δ -functional perturbation. We will call it the Green function of n^{th} order. The link between this Green function and the corresponding reflection coefficient (transfer function) is given by the inverse Fourier transform (7). The time response of a single Bragg mirror to a δ -function perturbation is the inverse Fourier transform of the reflectivity $r_B(\omega)$. For a uniform grating, the Green function of order 1 corresponds roughly to a square which averages the signal over a time inversely proportional to the Bragg bandwidth; the function of second order is a triangle function and so on [24].

Note that it is possible to introduce a Green function for the whole EC system in two ways, firstly as the sum of n Green functions $G_f(x) = \delta(x) + \sum_{n=1}^{\infty} G_n(x - n\tau)$, or secondly as the inverse Fourier transform of the reflectivity $f(\omega)$ of the extended cavity. Then the time response of the system on arbitrary perturbation is the convolution of the laser field and of the corresponding Green function $F(t) = G_f(t) \otimes E(t)$.

For usual parameters, the effective time response of the n th round trip to a δ -functional perturbation (8) is much lower (< 1 ns) than the feedback delay (10 ns). As a result, it is possible to carry out numerical integration only inside these effective time-intervals, which speeds up the calculation. Therefore we choose to use the set of Green functions $G_n(x)$ to calculate $F(t)$ instead of the single full Green function $G_f(x)$.

It is worth noting that in the limit of very low ($r_2 \gg r_B$) or high ($r_2 \ll r_B$) feedback level, the full Green function is mainly given by the first-order Green function. This means that the single round-trip approximation is correct in these cases while for other relations between r_2 and r_B , the use of a multi-pass approach is necessary. The first-order Green function is proportional, in the weak reflectivity approximation, to the apodization function η

$$\begin{aligned} G_1(t) &= -\frac{1-R_2}{r_2} \exp(-i\omega_0\tau + i\omega_B t) (\varrho L_B) \frac{1}{2T} \times \\ &\times \eta \left(\frac{L_B}{2} \left(\frac{t}{T} - 1 \right) \right) \left(\frac{1}{2} (\text{sign}[t] - \text{sign}[t - 2T]) \right) \end{aligned} \quad (10)$$

where $T = L_B n_B/c$.

In the following, we will consider system (1) with the basic parameters given in Table I. When the values of the parameters are changed or not shown in this table, they are given in figures. They are typical for FBG-EC lasers used as pump EDFA except that the bandwidth of FBG filters will be lower than usual. The large EC length (1 m) is obtained in practice by use of a fiber in order to avoid coherent effects in the feedback loop. The free spectral range of the EC is small in comparison with that of the IC. This discrepancy enables to distinguish, at weak feedback level, the different branches of steady-state solutions connected with different IC resonances.

III. ANALYTICAL APPROACH TO STEADY STATES

A. General Equations for Steady States

A steady state of the system (1) with fixed intensity I_s and frequency Ω_s is written as $E_s(t) = \sqrt{I_s} \exp(i\Omega_s t)$. The steady frequency Ω_s can be found from the following equation:

$$\{\Gamma G_N g(N_s, I_s) - x(\Omega_s)\} I_s = 0 \quad (11)$$

where I_s and N_s are functions of Ω_s with the following definitions:

$$\begin{aligned} N_s(\Omega_s) &= N_{\text{th}} + \frac{2}{\alpha \Gamma G_N} \left\{ \Omega_s - \frac{1}{\tau_{\text{in}}} \arg f(\Omega_s) - \Delta\omega_0 \right\} \\ I_s(\Omega_s) &= \frac{y(\Omega_s)}{x(\Omega_s)}, \\ y(\Omega_s) &= \Gamma \left(\frac{J}{eV} - \frac{N_s}{\tau_N(N_s)} \right) \\ x(\Omega_s) &= \frac{1}{\tau_{\text{ph}}} - \frac{2}{\tau_{\text{in}}} \ln |f(\Omega_s)| \equiv \frac{1}{\tau_{\text{ph}}^*} \end{aligned} \quad (12)$$

where τ_{ph}^* is the photon lifetime of the coupled cavity. The additional condition $y(\Omega_s) = 0$ gives a threshold current for each EC mode. When ϵ is equal to zero, the transcendental (11) is reduced to

$$(-\Delta\omega_0 + \Omega_s) \tau_{\text{in}} + \alpha \ln |f(\Omega_s)| - \arg f(\Omega_s) = 0. \quad (13)$$

In this equation, $\arg f$ is a multivalued function for which we will use the notation $\arg f = \text{Arg}f + 2\pi n$, where $-\pi < \text{Arg}f \leq \pi$. By adding $2\pi n$ to $\text{Arg}f$, we formally obtain frequencies Ω_s corresponding to other resonant frequencies of the IC. Therefore, we will call those steady states obtained at a given n “the n th IC branch”. They coincide with longitudinal modes of the solitary laser if its free spectral range $2\pi/\tau_{\text{in}}$ is considerably large in comparison to the second term in (13). But here we have to remember that system (1) enables only the study of the laser behavior involving only one of these longitudinal modes. In other words, the results which we obtained are correct as long as the field intensity and phase are changing slowly with respect to τ_{in} . In fact, in the case of strong feedback ($|r_B| \geq r_2$), the term $\alpha \ln |f(\Omega_s)|$ describing the frequency change induced by the phase-amplitude coupling becomes rather large. Therefore, we have to add $2\pi n$ to the multivalued function $\arg f$ with some nonzero integers n in order to compensate for this frequency shift and to maintain the system near the main longitudinal solitary laser mode with optical frequency ω_0 .

Note that in the numerical simulations of (1), we also need to pay special attention to the correct value of $\arg(F(t)/E(t))$ as mentioned in [23]. Its value has to be continuous because the phase of the field E and of the reinjected field $F(t)$ are continuous.

For a small Bragg reflectivity r_B , the well-known formula [32], [33] of the single-round-trip approximation is recovered from (13)

$$\Omega_s \tau_{\text{in}} + \frac{1 - r_2^2}{r_2} |r_B(\Omega_s)| \{ \alpha \cos \theta_s - \sin \theta_s \} = 0$$

where $\theta_s = (\Omega_s + \omega_0) \tau - \arg r_B(\Omega_s)$ ($\Delta\omega_0$ is omitted).

B. Influence of Compression Factor

The nonzero compression factor is known to modify the relaxation oscillation frequency through the Peterman factor. It is worth noting that nonzero compression ϵ is also one possible mechanism leading to a shift of the solitary laser frequency when the injection current is increased: $\Delta\omega_0^\epsilon = 2\pi k_\epsilon (J - J_{\text{th}})$. The corresponding rate is $k_\epsilon = (\alpha\epsilon\Gamma)/(4\pi e V)$. So the effective rate of frequency change, which could be derived from experiments, corresponds in our model to the sum: $k_{\text{eff}} = k_\epsilon + k$. This shift is easily obtained from the equation, which gives the frequency at steady states

$$\begin{aligned} ((-\Delta\omega_0 + \Omega_s) \tau_{\text{in}} - \arg f(\Omega_s)) (1 + \epsilon') + \alpha \ln |f(\Omega_s)| \\ = \alpha \epsilon' \frac{\Delta\mu}{2} \end{aligned} \quad (14)$$

where $\epsilon' = \epsilon/(G_N \tau_e)$ is the normalized saturation and $\Delta\mu = \Gamma G_N \tau_{\text{in}} \tau_e (J/(eV) - N_{\text{th}}/\tau_e)$ is the excess of the injection current above its threshold value. We have set $\tau_N(N_{\text{th}}) = \tau_e$.

For a nonzero compression factor, besides nonzero-intensity steady states which are given by (14), there are also trivial non-lasing steady states [see (11)] for which $I_s = 0$ and $N_s = N_{\text{th}} + \Delta\mu/(\tau_{\text{in}} \Gamma G_N)$. They are frequency equally spaced by $2\pi/\tau_{\text{in}}$ and steady-state frequencies can be found from the following equation:

$$(-\Delta\omega_0 + \Omega_s) \tau_{\text{in}} - \arg f(\Omega_s) = \alpha \frac{\Delta\mu}{2}. \quad (15)$$

Note that there is an infinite number of these trivial steady-state solutions even for weak feedback levels due to plural values of $\arg f$ [(13)] and the frequency-independent gain. They are saddle solutions belonging to different IC-branches and are unstable above threshold conditions ($\Delta\mu = -2 \ln |f(\Omega_s)|$).

In the next section, we will describe the structure of the steady states set given by (11)–(12). We will investigate the influence of different parameters such as the feedback strength, the intermediate reflectivity r_2 , the EC length L_{ext} , the external Bragg reflectivity r_B (or the coupling strength κ), the FBG bandwidth Δ , or the detuning between the FBG frequency ω_B and the solitary laser frequency ω_0 .

C. Frequency-Intensity Charts of Stationary Solutions

Fig. 2 shows the Bragg reflectivity and the steady states in the chart frequency intensity (Ω – I) for weak feedback ($r_2 \gg r_B$).

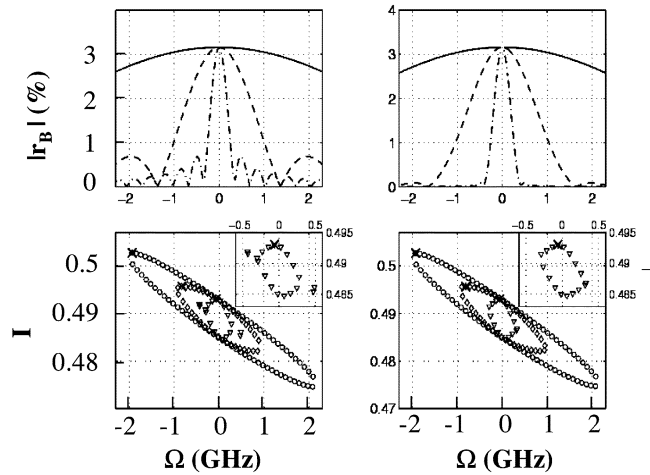


Fig. 2. Steady states in the frequency-intensity (Ω – I) domain (the lower panel). The left column corresponds to uniform FBG and the right column to cosine-apodized FBG. $L_{\text{ext}} = 1$ m, $\omega_0 \tau = 0 \pmod{2\pi}$, $R_2 = 0.1$, $\kappa = \sqrt{0.001}$, $\Delta\omega_0 = 0$, $\Delta\omega = 0$, $J = 70$ mA, $\epsilon = 1.6 \times 10^{-19}$ sm³. Bragg grating bandwidth $2\Delta = 0.1$ nm (circles), $2\Delta = 0.02$ nm (diamonds), and $2\Delta = 0.005$ nm (triangles). The curves in the upper panel give the corresponding frequency Bragg grating response $|r_B(\Omega)|$. Mark "x" locates maximum-gain mode.

The left column corresponds to the uniform grating and the right to the apodized case. Steady states lie on a closed curve (almost an ellipse). The upper branch of this curve consists of EC modes (nodes which can be seen as constructive interferences between the laser field and the delayed field due to the EC), and the lower branch consists of antimodes (destructive interferences). They appear in pairs via a saddle-node bifurcation when a control parameter is changed. Such bifurcations can be seen in the next section, when the solitary laser frequency is increased. Antimodes are saddle-fixed points and they are always unstable. EC modes appear as stable nodes but they can lose stability through a Hopf bifurcation. For a rather wide bandwidth of the Bragg reflector $\Delta \gg \Omega^* = (1 - R_2)(r_B^{\text{max}}/r_2)(1/\tau_{\text{in}})\sqrt{1 + \alpha^2}$ (here $r_B^{\text{max}} = r_B(\omega_B)$), steady-states frequencies lie in the limits $-\Omega^* \leq \Omega_s \leq \Omega^*$. This case corresponds totally to the usual results of external feedback without filtering. Free spectral range between EC modes is equal approximately to $2\pi/\tau$. The so-called maximum gain mode (mode with highest power) [33] is located for rather wide bandwidth on the left (red) side of the ellipse (or more precisely of the zero IC branch). It is depicted by "x" in the figure. It was shown [33] that there is always a feedback phase condition for which the maximum gain mode is stable at arbitrary feedback strength. In other words, this mode is not always stable [2], [34]–[36]. In our calculations, the maximum gain mode was checked to be numerically stable except for a few simulations in the region $R_2 \approx R_B$. A more detailed stability analysis will be necessary but is beyond the scope of this study.

A decrease of the FBG bandwidth, results in a decrease of the number of modes, in the shrinking of the ellipse curve and in the distortion of its form (oval-like). Moreover, the maximum gain mode is pulled toward the Bragg frequency as shown in Fig. 2. Similar results were obtained in [32] for a single Lorentzian filter placed into an EC and when only a single round trip was taken into account. In contrast to the case of the Lorentzian filter

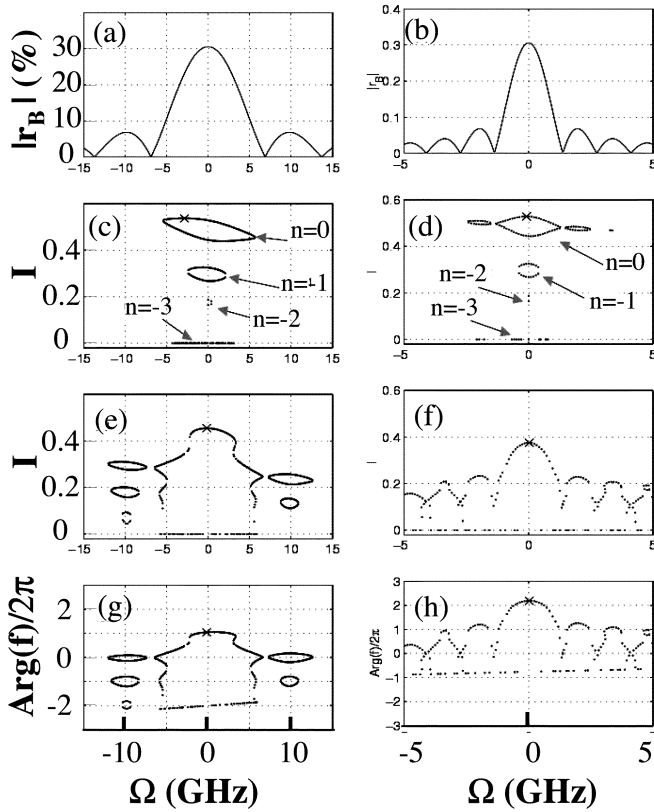


Fig. 3. Steady states for strong feedback ($\kappa = \sqrt{0.1}$) and for uniform FBG. The left column corresponds to a Bragg grating bandwidth equal to $2\Delta = 0.1$ nm, and the right to $-2\Delta = 0.02$ nm. (a) and (b) represent FBG reflectivity. (c) and (d) $R_2 = 0.1$. (e) $R_2 = 0.005$. (f) $R_2 = 0.0001$. (g) and (h) are the corresponding phase relation ($\arg(f(\Omega_s))$) versus frequency (Ω_s) for figures (e) and (f). The other parameters are the same as in Fig. 2.

[32], for a very narrow FBG bandwidth and moderate feedback level, the shape of the steady-states curve looks like a convex rhomb instead of a concave rhomb. There is also the appearance of satellite branches corresponding to side lobe in the response of the FBG (see the insert on the left column of Fig. 2). The satellites shown in the curve with triangles are obtained for $r_B^{\max}/r_B^{\text{sidelobe-max}} \approx 4.6$ and $\Omega^*/\Delta \approx 6$. Note that we consider here a long EC ($L_{\text{ext}} = 1$ m) and that $\Delta\Omega_{\text{ext}} \approx 0.1$ GHz is three times narrower than the FBG bandwidth Δ . Increasing the delay leads to an increase of the number of modes which cover the same curve.

We also investigate the influence of FBG apodization on the laser characteristics. An example of cosine-apodized FBG is given on the right column of Fig. 2 for the same reflectivity maximum and the same half-width-at-half-maximum. It illustrates the fact that apodization suppresses the satellites. This behavior is directly connected to the existence of side lobes in the reflectivity. Gaussian apodization brings similar results.

For previous parameters, all the considered steady states have a value of $|\arg f| < \pi$ and are located near the zero IC resonance. When the feedback level becomes stronger ($r_2 \simeq r_B^{\max}$), for instance in the case of antireflection coating on the coupling face, the pattern of modes disposition is qualitatively changed [Fig. 3(c)]. The steady-states curve expands along the frequency axis and becomes more similar to an oval. For these parameters, we obtain several branches of solutions (several oval-like

curves) located one under the other in the planes (Ω – I) approximately at the same frequencies. Each curve corresponds to a different IC branch (a different value of n). Note that IC branch with $n = -3$ represents trivial nonlasing steady states [see (15)].

In Fig. 3(d), the main axis [whose angle is given by $\arctan(2/\alpha)$] is not preserved contrary to what is observed in Fig. 2. In other words, with the actual values of parameters, dispersion compensation (or control of the chirp due to the α parameter), is more pronounced for high feedback strength (or low reflectivity r_2). If the FBG bandwidth becomes sufficiently narrow so that reflectivity lobes are within the frequency interval $[-\Omega^*, \Omega^*]$, satellite branches can appear [Fig. 3(d)]. These satellites can consist of a few IC branches with different n [Fig. 3(e)]. This property corresponds to the fact that one can obtain a “zero” frequency with a nonfundamental IC mode and several round trips inside the EC. Apodization of the FBG leads again to the suppression of satellites.

When the feedback strength is further increased ($r_2 < r_B^{\max}$ is decreased at fixed $r_B^{\max}$), we first observe in the lower part of the chart, the collision of two adjacent ellipses: it starts out with ellipses corresponding to lower intensities while ellipses with higher intensities are almost unaffected. After a collision, a new curve is formed, consisting of the top of an ellipse (almost unchanged) and of two accompanying arms. They are the reminiscence of the “lower” ellipse and of the lower part of the “highest” ellipse. The situation presented in Fig. 3(e) (with no satellites) is the first one encountered. Increasing the feedback level propagates the collision among the satellites [Fig. 3(f)] and then with the branch corresponding to the main Bragg maximum.

In summary, when the feedback strength is increased, all curves merge into one while the frequency of the maximum-gain mode moves toward the centered frequency of the filter [Fig. 3(f)]. Obviously, this collision is due to saddle-node bifurcations under which EC modes (nodes) of lower curve and antinodes (saddles) of upper curve disappear in pairs. After a collision, the resulting curve consists of nodes in the upper part and trivial nonlasing EC steady states (saddles) with zero intensity. The dominant branch to which the mode with maximum gain belongs, has $n \neq 0$ [Fig. 3(g) and (h)]. This collision leads to the appearance of frequency intervals in which there are no more antinodes.

When we slightly shift the FBG frequency (ω_B) from zero (solitary laser frequency), the steady-states curves are correspondingly shifted. Increasing the detuning leads to the appearance of two steady-states curves localized near the Bragg and free-laser frequencies as for the Lorentzian filter [32]. The difference is that for a uniform FBG the curve localized near zero frequency is the satellite associated with the first side-maximum of the FBG. For larger detuning, more satellites may exist (Fig. 4). On the contrary, when the FBG is apodized or when the lobes in the reflectivity response are suppressed, this effect disappears and the EC modes are located only around the main maximum. Then only the mode located closest to the solitary laser frequency is maintained (Fig. 4, right column). It clearly shows the importance of apodization in order to avoid competition between the modes favored by the maximum of the

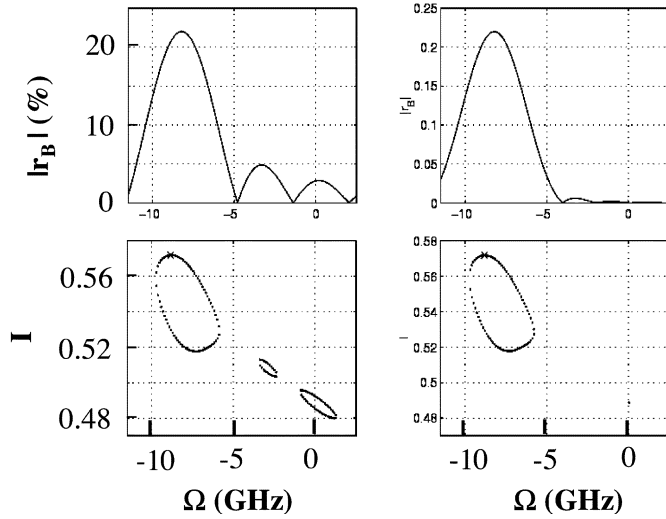


Fig. 4. Steady states in the domain frequency-intensity (Ω - I) (the lower panel). The left column corresponds to uniform FBG and the right column to cosine-apodized FBG. $\kappa = \sqrt{0.05}$, $\Delta\omega = 0.06$ nm. FBG bandwidth $2\Delta = 0.05$ nm. The curve in the upper panel gives the corresponding frequency FBG responses $|r_B(\Omega)|$. Mark "x" locates maximum-gain mode. The other parameters are the same as in Fig. 2.

gain curve and the ones favored by the maximum reflectivity of the filter. The steady-states curve corresponding to the main maximum is the same as for the uniform FBG, and the same remarks as previously made hold when the feedback strength is increased (collision of branches and spread of the modes over the frequency axis).

IV. L - I CHARACTERISTICS

To obtain L - I curves, we use numerical simulations. Initial conditions are chosen as those of the mode with maximum gain at a low value of the injection current. So, the initial conditions are taken for an EC mode associated to the main IC resonance. Then the current is slowly increased and $\arg(F/E)$ is kept continuous as a function of time [23]. The real temporal evolution of the system is averaged over $\delta t = 10$ ns. A point on the averaged trajectory coincides with a steady state if the steady state (or attractor in its vicinity) is stable. Another way to proceed is to increase the injection current step by step and to wait after each step for transient dumping, which requires longer computation time.

In Figs. 5–7 (lower panel), L - I curves are depicted for different feedback levels of uniform FBG. One can see that their shape is strongly affected by the feedback strength. To better understand the reasons explaining the shape of a L - I curve, let us consider the dependence of steady states on the pump current [or on the solitary laser frequency which is proportional to it in accordance with formula (14)]. Corresponding results are presented in the upper panel of Figs. 5–7 in the plane (solitary laser frequency) — (steady state frequency). In the same figures, numerical averaged system trajectories are shown when the current is temporally changed. To depict analytical and numerical ($\Delta\omega_0$ - Ω) curves in this two-dimensional (2-D) space (the third dimension is the injection current), we do not consider the compression gain factor (or the term $\alpha\epsilon'\Delta\mu/2$ from (14)) for the analytical curves, and ignore equivalent frequency shift for the numerical curves.

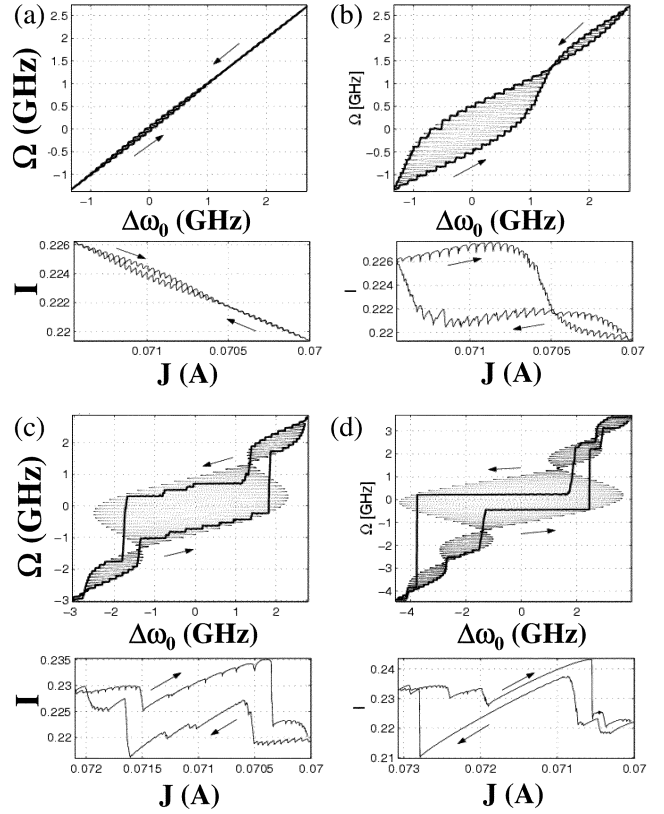


Fig. 5. Steady states and path obtained by numerical integration for uniform FBG in the ($\Delta\omega_0$ - Ω)-domain (upper panel); corresponding L - I curves (lower panel). FBG bandwidth $2\Delta = 0.02$ nm, $\epsilon = 1.6 \times 10^{-16}$ sm³, $k = -3$ THz/A. (a) $\kappa = \sqrt{10^{-6}}$. (b) $\sqrt{10^{-4}}$. (c) $\sqrt{10^{-3}}$. (d) $\sqrt{3} \times 10^{-3}$. The other parameters are the same as in Fig. 2.

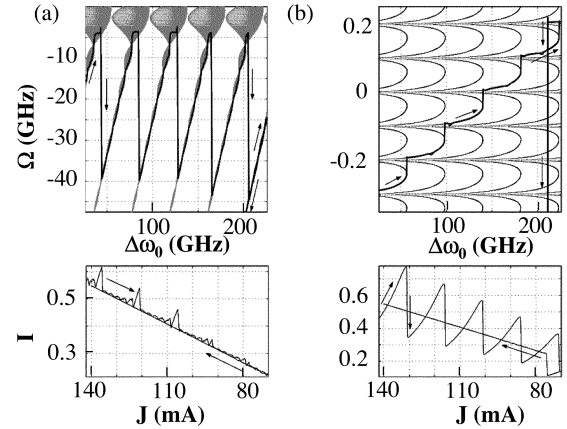


Fig. 6. Same representation as in Fig. 5 for FBG bandwidth $2\Delta = 0.1$ nm. (a) $\kappa = \sqrt{0.05}$. (b) $\sqrt{0.1}$. The other parameters are the same as in Fig. 5.

Below, we will distinguish three main intervals of feedback strength with qualitatively different steady state characteristics and behaviors, though there can be strong changes in the laser characteristics inside each interval.

A. Coupled-Cavity Regime: Single Internal-Cavity Mode Operation

For the first interval of feedback depths ($R_B^{\max}/R_2 \leq 0.1$), the system maintained its operating points on one longitudinal mode of the solitary laser under a current modulation. Fig. 5 illustrates the change of the coupled laser frequency and inten-

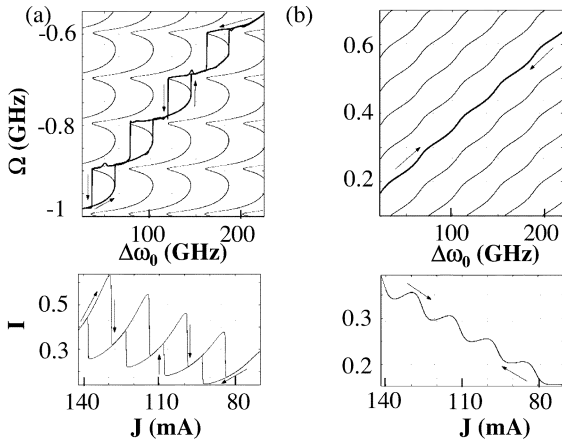


Fig. 7. Same representation as in Figs. 5 and 6 for FBG bandwidth $2\Delta = 0.1$ nm, $\kappa = \sqrt{0.1}$. (a) $R_2 = 0.05$. (b) 10^{-3} . The other parameters are the same as in Fig. 5.

sity when the bias current or the phase is varied. In the upper panel of the figure, only the *zero* IC branch of steady state solutions [(14)] is shown. In the case of conventional feedback, the stationary frequency describes a pure sinusoidal shape along a straight line when the free-laser frequency is varied, while here the figure reveals the response of the filter. The conventional response given by the EC mode (antimode) structure is attenuated by the finite optical bandwidth of the reflectivity. As was mentioned in Section III, there are also other branches due to other internal cavity resonances, which correspond to other longitudinal modes of the solitary laser at the feedback level under consideration ($\alpha |\ln |f||_{\max} \ll 2\pi$). These branches are separated approximately by $2\pi/\tau_{\text{in}}$ in the frequency $\Delta\omega_0$ axis and have similar shapes. Numerical simulations show that if the system is in a vicinity of an IC branch (longitudinal mode), it moves along the same branch whatever the changes in the frequency $\Delta\omega_0$ for this regime. Note that our model does not use a frequency-dependent gain so that here we assume that the changes in $\Delta\omega_0$ are considerably less than the gain linewidth. We check effectively that within the framework of this model, the EC laser regimes are given in a small frequency-interval such that $\Delta\omega_0 \ll 2\pi/\tau_{\text{in}}$.

Approximately the same feedback levels were considered in detail for the Lorentzian filter [37] in the Lang–Kobayashi approximation [1]. We have obtained similar results.

- 1) At almost zero feedback level, the system moves along a single-valued steady-state curve.
- 2) At low feedback, an increase (decrease) of $\Delta\omega_0$ leads to jumps of the system from one EC mode to an adjacent mode due to annihilation of a pair of steady states in a saddle-node bifurcation.
- 3) At a higher feedback level, there are jumps equivalent to several EC-mode spacings when the laser operates near the central frequency of the filter.

Fig. 5(a)–(d) illustrates such results and show clearly the evolution of the frequency of the laser when the optical phase is changed (through the bias current or the solitary laser frequency). This behavior indicates the path among the EC modes which can belong to different satellites. For instance, Fig. 5(a) corresponds to the case 2) for the mode at the frequency (0

GHz) of the FBG main maximum and to the case 1) for the mode at the frequency (2 GHz) of a side-lobe maximum, while Fig. 5(b) displays the behaviors 2) and 3) for the side and central maxima correspondingly. Hysteresis phenomena involving two EC modes or all EC modes belonging to one of the FBG maxima are evident in both the frequency domain (the upper panel) and L – I dependence (the lower panel). The corresponding L – I curves are slightly modulated with a period, which is approximately determined by the effective rate k_{eff} . The frequency associated to this period is linked to the distance between two extrema of $\ln |f|$ [(14)] and is given by $1/(k_{\text{eff}}\tau)$. In Fig. 5(b), bends appear when the system switches from one mode to the next one.

In addition to the above regimes, we observed, with an increase of $R_B^{\max}$, switchings between the main and side maxima of FBG as shown in Fig. 5(c) and (d). They correspond in Fig. 3(d) to jumps from the main steady-states oval-like curve to its satellite. When the system moves inside the main lobe, the laser frequency gets closer to the central frequency when the feedback strength is increased [compare Fig. 5(c) and (d)]. The paths obtained for a decrease and an increase of $\Delta\omega_0$ demonstrate hysteresis phenomena. It is worth noting that there is a coexistence of stable EC modes for the central and side maxima. Small variations of $\Delta\omega_0$ lead to abrupt changes in the L – I characteristics for these parameters. They correspond to switchings between maxima of the FBG reflectivity.

The track of the real temporal evolution on a time scale much lower than the relaxation rate of the system (~ 1 ns $^{-1}$) shows that attractors along which the system moves are stable steady states. When switching points do not coincide with saddle-node bifurcation points, path are cycles or more complex attractors.

B. Confused Cavity Regime: Competition Between IC Modes

For stronger feedback depth ($0.1 < R_B/R_2 \leq 1$), switchings between different longitudinal (IC) modes can appear.

We noticed that when the feedback strength is increased, the value of the laser frequency is within a larger interval, so that hopping between EC modes shows a bigger change in frequency [compare Fig. 5(a) to (d)]. For a higher feedback level, the frequency distance between EC modes belonging to different IC branches reduces as shown in the upper panel of Fig. 6(a). It follows that when $\Delta\omega_0$ is increased in a wide interval (over a few IC-branch spacings $2\pi/\tau_{\text{in}}$), jumps between IC branches are observed. They correspond to hopping between IC modes of the solitary laser. Therefore, the laser frequency (Ω) changes abruptly (~ 45 GHz $\approx 2\pi/\tau_{\text{in}}$, compared to Fig. 5) and the system operates far from the transmission band of the filter. Then, the frequency is shifted back to the initial region and the process is repeated. The averaged L – I characteristic (Fig. 6(a), lower panel) displays this sudden hopping. In the framework of system (1), the laser operation is maintained in a “single-longitudinal IC-mode” operation. Obviously, the mode hopping demonstrates that beating between IC longitudinal modes is possible at these parameters. That is why, in this case, multimode equations with frequency dependent gain and losses have to be introduced if we want to correctly describe the laser behavior.

Note that when $\Delta\omega_0$ is decreased the system moves along only one IC branch.

When R_B is of the order of R_2 , $\alpha |\ln |f||_{\max}$ becomes comparable with 2π and the EC structures belonging to adjacent IC branches attract each other and interpenetrate into each other. Fig. 6(b) illustrates this change of behavior. Note the difference in scale with Fig. 6(a) and the collisions between IC branches. A consequence is that one can find solutions with different values of n at a fixed frequency as in Fig. 3(c) (note that here frequencies of trivial nonlasing steady states are not shown).

Contrary to the previous case, for this second regime, the concept of IC modes loses its sense and one cannot consider a single-frequency IC coupled to an EC. Under such conditions, another kind of switching between adjoining IC branches appears. When the frequency $\Delta\omega_0$ enlarges from its near-zero value, the system firstly moves along a stable EC mode belonging to the zero IC branch and then jumps through a saddle-node bifurcation to the nearest stable EC mode belonging to the next branch and so on (Fig. 6(b), upper panel). In this case frequency jumps are approximately equal to EC-mode spacings (~ 100 MHz, the same spacing as in Fig. 5). These jumps are smaller than in Fig. 6(a), but they result in stronger modifications of the L - I characteristics [lower panel of Fig. 6(b)]. Such behavior is closer to the one of the third regime considered below. Nevertheless, when we decrease $\Delta\omega_0$, the system is not locked anymore by the nearest EC mode belonging to the adjacent IC branch. Then one observes a jump to another IC branch, far from the transmission band of the filter. The change in Ω is then approximately the IC free spectral range [as in Fig. 6(a)].

C. Long Cavity Regime

For very strong feedback levels ($R_B/R_2 > 1$), the IC plays a minor role. A new EC-mode structure appears and the operating frequency becomes more stable.

To consider this case with $R_B > R_2$, R_2 is decreased while R_B is set at 0.1. It is achieved in practice by antireflection coating. First, the interpenetration of adjacent IC branches enlarges, as shown in Fig. 6(b). The shape of IC branches becomes more asymmetric and elongated. The EC-modes structure contains very narrow peaks which are directed to the left along $\Delta\omega_0$ -axis. In accordance with (14), these peaks correspond to minima (stopbands) of the reflectivity $|f(\Omega)|$ of the external interferometer (note that, in this case, $f(\Omega)$ can be approximated by: $f(\Omega) \simeq 1 + (1 - R_2)(r_B/r_2) \exp(-i(\Omega + \omega_0)\tau)$). It occurs when the minima are rather close to zero (for $(1 - R_2)r_B/r_2 \simeq 1$); $\ln |f(\Omega)|$ in the function $\Delta\omega_0(\Omega)$ gives sharp negative deeps separated by $2\pi/\tau$. The separation between two adjacent maxima of the same IC branch (or two points of different IC branches, at the same Ω) is constant in the $\Delta\omega_0$ -direction and is approximately equal to $2\pi/\tau_{\text{in}}$.

When r_2 is further decreased, a critical point is obtained for which $|f(\Omega)|$ tends to zero, and $\ln |f(\Omega)| \rightarrow -\infty$ that leads to infinite-long peaks. The behavior of $\arg f(\Omega)$ undergoes dramatic changes. $f(\Omega)$ is represented by a circle-like curve in the complex plane ($\text{Re}f, \text{Im}f$). Before reaching this critical point, this function lies completely in the right half-plane and is symmetric with respect to the real axis. $\arg f(\Omega)$ is lying in the in-

terval $(-\pi/2, \pi/2)$ when Ω is varied. The appearance of a peak (or a long tail) in the $(\Delta\omega_0, \Omega)$ -plane corresponds to the fact that the circle-like curve gets closer to zero in the complex plane. In the vicinity of the peak, $\arg f$ is approximatively equal to zero and is slightly changed. A critical point appears when the circle-like curve touches zero on the imaginary axis. Increasing again the feedback strength makes this curve intersect the imaginary-axis. Then $\arg f(\Omega)$ is lying in the interval $[-\pi, \pi]$ and $\arg f$ is strongly changed (over more than π) close to the peak point. It results in the effective merging of adjacent IC-branches with different n . More precisely every EC antimode with a given n joins to an EC mode associated to $n + 1$. This process begins in the center of the FBG reflectivity, and it can be understood by looking at Fig. 6(a) ($\Omega = 0$). This phenomenon spreads to the region of smaller frequency values with decreasing R_2 . As a final result, the average slope of the new joined continuous branch in the $(\Delta\omega_0, \Omega)$ -domain [Fig. 7(b)] becomes equal to $(1/\tau)/(1/\tau_{\text{in}}) \simeq 0.0024$ while the slope of a continuous branch was 1 in the case of weak feedback. When R_2 is further decreased, these branches hold saddle-node bifurcation points as above and the system path looks like the one presented in Fig. 5(a): Ω switched by an EC-mode spacing ($2\pi/\tau$), except that here there is a jump to another IC branch which results in strongly modulated L - I curve. The modulation period is given by about $1/(k_{\text{eff}}\tau_{\text{in}})$ [Fig. 7(a)].

When the feedback level is further increased, the joined branches become smoother [Fig. 7(b)]. In fact, they correspond to new longitudinal modes of the long-laser cavity created by the mirrors R_1 and R_B . Internal interferometer consisting of the antireflection coated facet and of the mirror R_1 only slightly modulates every branch. The system path is without jumps and hysteresis [upper panel of Fig. 7(b)]. The L - I curve displays “kinks,” which become smaller when R_2 decreases [the lower panel of Fig. 7(b)]. Disappearance of saddle-node bifurcations in $(\Delta\omega_0 - \Omega)$ -domain is due to collisions between adjacent IC branches [remember the ellipses shown in Fig. 3(e)]. Note that when $r_2 = 0$, $r_2 f(\omega) = r_B(\omega) \exp(-i(\omega + \omega_0)\tau)$, so that when $r_2 \rightarrow 0$, there is no singularity as one has strictly to consider $\ln(r_2 f(\Omega))$ instead of $\ln(f(\Omega))$. Within this limit, the reflectivity of the coupled cavity is that of the Bragg gratings, the laser is operating on the long cavity and kinks disappear.

Almost all the properties described above are kept for apodized FBG. The main difference is that the absence of side-lobe in the reflectivity suppresses jump between maxima of reflectivity. This results in behavior similar to that presented in Fig. 5(a) for a low feedback level.

V. RIN CHARACTERISTICS

To calculate the RIN, we have to take into account stochastic noise terms that are usually introduced in the (1) in the real form

$$\begin{aligned} \dot{I} &= \left[\Gamma G_N g(N, I) - \frac{1}{\tau_{\text{ph}}} + \frac{2}{\tau_{\text{in}}} \ln \left| \frac{F}{E} \right| \right] I + R + F_I(t) \\ \dot{\varphi} &= \frac{1}{2} \alpha \Gamma G_N (N - N_{\text{th}}) + \frac{1}{\tau_{\text{in}}} \arg \left(\frac{F}{E} \right) + \Delta\omega_0 + F_{\varphi}(t) \\ \dot{N} &= \frac{J}{eV} - \frac{N}{\tau_e} - G_N g(N, I) I + F_n(t). \end{aligned}$$

Here, $F_I(t)$, $F_\varphi(t)$, $F_n(t)$ are the Langevin noise sources associated with intensity, phase, and carrier density. These Langevin forces satisfy the following relations [38], [39]: $\langle F_i(t) \rangle = 0$, $\langle F_i(t) F_j(t') \rangle = 2D_{ij}\delta(t - t')$, where D_{ij} are diffusion coefficients

$$\begin{aligned} 2D_{II} &= 2RI, \quad 2D_{\varphi\varphi} = \frac{R}{(2I)} \\ 2D_{NN} &= 2(RI + S) \\ 2D_{IN} &= -2RI, \quad 2D_{I\varphi} = 2D_{\varphi N} = 0. \end{aligned}$$

The spontaneous emission term is $S = N/\tau_e$. As shown in [23] and [40], EC influences the spontaneous emission rate R which becomes dependent on the external reflection and therefore on the frequency. We use the next formula obtained for an arbitrary amount of feedback [23]

$$R = \frac{n_{sp}}{\tau_{ph}} \frac{\alpha_{in} + \frac{1}{L_D} \ln \left(\frac{1}{(r_1 r_2 |f(\omega)|)} \right)}{\alpha_{in} + \frac{1}{L_D} \ln \left(\frac{1}{(r_1 r_2)} \right)} \times \left[\frac{(r_1 + r_2 |f(\omega)|) (1 - r_1 r_2 |f(\omega)|)}{2r_1 r_2 |f(\omega)| \ln \left(\frac{1}{(r_1 r_2 |f(\omega)|)} \right)} \right]^2. \quad (16)$$

In numerical simulations, these Langevin noise forces can be represented as

$$\begin{aligned} F_I &= \sqrt{\frac{2I(t_i)}{\Delta t}} R \chi_I, \quad F_\varphi = \frac{1}{I} \sqrt{\frac{I(t_i)}{2\Delta t}} R \chi_\varphi \\ F_N &= \sqrt{\frac{2N(t_i)}{\tau_e \Delta t}} \chi_N - \sqrt{\frac{2I}{\Delta t}} R \chi_I \end{aligned}$$

where χ_I , χ_φ , and χ_N are Gaussian random variables with zero mean and unity standard deviation ($\langle \chi_i^2 \rangle = 1$). Δt is the time interval over which the noise is held constant in numerical simulations ($\Delta t = 1$ ps).

Analogous Langevin noise term for field equation in (1) is $F_E = (\sqrt{R/\Delta t}) \chi_E$ and $F_N = -\sqrt{R/\Delta t} (E^* \chi_E + E \chi_E^*) + (\sqrt{2N/\tau_e \Delta t}) \chi_N$, $\chi_E = 1/\sqrt{2} (\chi_E' + i\chi_E'')$, where χ_E' , χ_E'' , χ_N are Gaussian variables with zero mean and unity standard deviation.

RIN is defined as the square of the ratio between the standard deviation of fluctuations of the optical power $\delta I(t)$ and the mean power $\langle I \rangle$: $RIN = \langle \delta I^2 \rangle / \langle I \rangle^2$. RIN(ω) can also be expressed at a given frequency as $RIN = \langle |\delta I(\omega)|^2 \rangle / \langle I \rangle^2$, where $\langle |\delta I(\omega)|^2 \rangle = \lim_{t \rightarrow \infty} 1/t \left| \int_0^t \delta I(t') \exp(-i\omega t') dt' \right|^2$.

To calculate this power spectral density, the Fourier transform is applied to data samples. Several spectra are averaged to reduce statistical variance. The sampling rate is 20 ps, which gives a frequency range of 25 GHz.

Calculation for RIN has been made by averaging over time or over frequency. We stress here that this averaging explained the quite high RIN-value. In Fig. 8, we present an example of the dependence of averaged RIN as a function of feedback level where the average has been made over frequency for two values of the gain compression factor ϵ . Frequency-dependent RIN were presented in our previous work [24] demonstrating numerous peaks

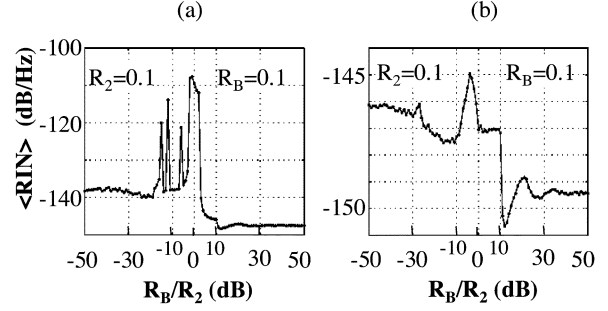


Fig. 8. Averaged (over frequency) RIN versus feedback depth. Uniform FBG bandwidth $2\Delta = 0.1$ nm. (a) $\epsilon = 1.6 \times 10^{-19} \text{ sm}^3$, $\Delta\omega_0 = 0$. (b) $\epsilon = 1.6 \times 10^{-16} \text{ sm}^3$, $\Delta\omega_0 = -10$ GHz. In the left half of the figures, the coupling reflectivity of the laser R_2 is kept constant to 0.1, while in the right half the figures, the Bragg reflectivity R_B is equal to 0.1. The other parameters are the same as in Fig. 2.

due to beatings between EC modes. For very small ϵ [Fig. 8(a)], it can be clearly seen that the RIN is greatly influenced by the feedback rate. We can distinguish three regions as in Section IV for L - I dependence. The first one extends from -50 to -20 dB for the ratio between the Bragg reflectivity and the reflectivity of the laser coupling facet. In our case of long cavity the averaged RIN is approximately constant. The laser operates in the vicinity of the longitudinal mode of the solitary laser. Due to that, RIN characteristic is closed to its value for the solitary laser. The second domain extending from 5 to 50 dB is characterized by a low averaged RIN, where the control parameters R_2 is varied and R_B is maintained constant. It corresponds to strong feedback regime when the system operates in the long cavity regime. Change of losses for this cavity explains the difference in RIN between the left and right parts of Fig. 8(a) and (b). The electrical spectrum of RIN calculated at a minimum of average RIN demonstrates in the full frequency range 0 – 25 -GHz EC beating peaks [24] with a classical enhancement at the relaxation frequency. Simulations show that the RIN at the relaxation frequency can be diminished for narrower filter.

When the gain compression factor ϵ is large, it can stabilize the system as shown by the difference between Fig. 8(a) and (b). The laser operates near the zero IC resonance at any feedback level and the RIN is not strongly modified, though one can observe a slight increase when $R_2 \approx R_B$.

An intermediate domain (ID) reveals a dramatic enhancement of the averaged RIN. It is mainly due to operation in the second regime, i.e., mode hopping between the IC resonances. Examples of such a behavior are presented in Fig. 9. Fig. 9(a) corresponds to the left side of the ID (R_B is slightly smaller than R_2). During the transient period the system spends considerably long time near the zero IC resonance ($n = 0$), then goes to the adjacent IC resonance and comes back. This process is repeated until the system maintains its operation on the stable adjacent IC resonance. We want to stress again that for this second regime a multimode model is required to ensure that the results are correct and that no behavior involving a few IC modes appear at these parameters. On the right side of the ID (R_B is slightly greater than R_2), the system remains in a vicinity of the zero IC resonance demonstrating complex quasiregular pulsations [Fig. 9(b)]. Recall that higher gain compression suppresses IC modes jump.

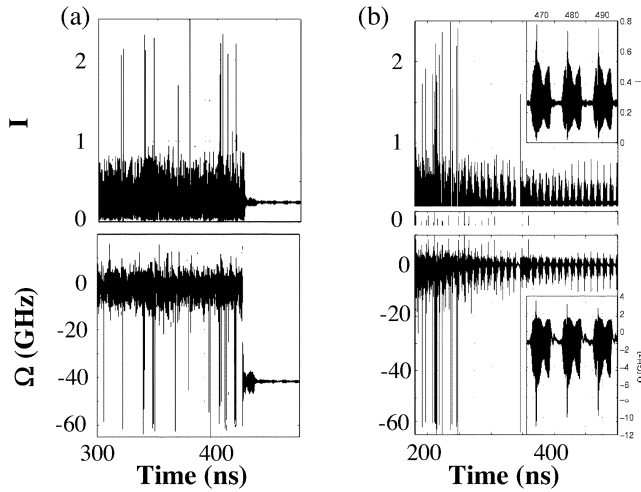


Fig. 9. Time evolution of the instant laser frequency (upper panel) and the laser intensity (lower panel). (a) $R_B/R_2 = 10^{-0.2} \approx 0.0631$ ($R_B = 10^{-1.2} \approx 0.0631$, $R_2 = 0.1$). (b) $10^{0.1} \approx 1.259$ ($R_B = 0.1$, $R_2 = 10^{-1.1} \approx 0.0794$). The other parameters are the same as in Fig. 9(a).

VI. DISCUSSIONS AND CONCLUSION

In this paper, we present a method to analyze the dynamics of semiconductor lasers coupled to any kind of filtering feedback with arbitrary strength. For a correct consideration of strong feedback, multiple reflections in the EC are taken into account. They are introduced with the help of Green functions, which permits to speed up numerical calculations and to describe strong optical feedback.

Analysis of the stationary solutions enables us to distinguish three intervals of feedback strength with qualitatively different steady-state sets.

The first interval is defined by $\alpha |\ln |f||_{\max} \ll 2\pi$, where f is related to the reflectivity of the coupled cavity. This coupled-cavity regime is characterized by the existence of EC modes (antimodes) located near resonance frequencies of the internal cavity ($\Delta\omega_0 \approx 2\pi n/\tau_{\text{in}}$).

In the second interval ($\alpha |\ln |f||_{\max} \approx 2\pi$), new EC modes appear in the intervals between adjacent IC resonances when the feedback strength is increased, and one reaches a situation where all the EC modes are covering the whole interval separating two IC modes. Then with a further increase of feedback level, an overlap of domains between different IC resonances occur and the concept of IC modes loses sense. In this confused-cavity regime, none of the two cavities is predominant.

For the third interval ($\alpha |\ln |f||_{\max} > 2\pi$), the EC is the laser cavity and the EC modes are the longitudinal modes of this coupled laser. This modification of the steady state set is accompanied by the appearance of new solutions and the collapse of old ones in the plane frequency-gain of the ellipse which usually contains the modes and so-called antimodes.

L - I dependencies are explained using the above analysis of steady states. For the first interval of feedback levels, the laser is operating near one IC resonance (longitudinal mode of the solitary laser) and the L - I curve is slightly modulated. The modulation period is determined by $1/(k_{\text{eff}}\tau)$. In the second regime, when the pump current or the phase is varied, there is IC mode

hopping and a modulation in the L - I curve. This modulation is characterized by a period closed to $1/(k_{\text{eff}}\tau_{\text{in}})$ and by an amplitude, which reduces when R_2 is decreased. The average slope of L - I curve becomes smaller. In the third regime the system can maintain its operation near the central FBG frequency under pump modulation in a wide domain. All these regimes have been analyzed with respect to a chirp $\Delta\omega_0$ which is equivalent to a detuned filter.

Finally, RIN is given as a function of feedback strength for a large variation (100 dB) of the ratio between the reflectivity of the laser coupling face R_2 and the Bragg reflectivity R_B . RIN is analyzed with respect to the three different regimes. The second regime is characterized by a RIN increase as it could be expected from the appearance of jumps, while the third regime offers the lowest RIN due to the suppression of the internal cavity and hence of the competition between the two coupled cavities.

Some properties are similar to the ones obtained for a Lorentzian filter [32]. The main difference is related to the existence of side lobes in the FBG reflectivity which allows switchings between these maxima. In both situations many properties can be described on the basis of the Lang-Kobayashi approximation. Our model suggests the roles of other internal cavity modes by the insertion of a feedback term (logarithm) which has two contributions: one for the losses, modification expressed by the modulus of the effective reflectivity and the other one for the additional phase, brought by the phase of the same effective reflectivity. Obviously, the limits of this model are attained in the second regime when several IC modes have an interplay in the dynamics. This can lead to multimode pulsations as observed experimentally [41]–[43]. A multimode model for the solitary laser is then required. A first answer has been given to describe static properties through a transfer function [44]. The multimode extension of the Lang-Kobayashi equations were used to explain such experimental results as in-phase and out-of-phase multimode dynamics and IC mode-beating [27], [28]. Note that jumps between IC modes, obtained in the present work, recall some peculiarities of multimode behaviors observed experimentally [41], though our model describes a single-longitudinal operation. The next step can be an extension of the model to a multimode one. It will enable us to understand the transition from a single-longitudinal mode operation to a multimode behavior. In fact, in a multimode approach a frequency dependent gain has to be taken into account when frequency shift is large in comparison to the gain linewidth. Note that in our results frequency jumps and Bragg gratings bandwidth are comparatively small.

In addition to this modification to the usual Lang-Kobayashi model, we have shown that multiple round trip could be easily accounted for. This multiple round-trip approach has been included systematically in the linear analysis and in the numerical simulations. Obviously, the influence of such terms on dynamics deserves a more systematic study that is beyond the scope of this paper. The same remarks hold for a detailed stability analysis of such system.

This kind of study is important for sources in optical telecommunications where there are several examples of laser diodes coupled to FBGs, which becomes a key component. These results and conclusions encourage us to pursue this work.

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